of common objects like circle, right circular cone, right circular cylinder, sphere can be expressed as a function with one or more variables.

In class IX, we had studied the concept of sets. We have also seen how to form new sets from the given sets by taking union, intersection and complementation.

Now we are about to study a new set called "cartesian product" for the given sets A and B.

1.2 Ordered Pair

Observe the seating plan in an auditorium (Fig.1.1). To help orderly occupation of seats, tokens with numbers such as (1,5), (7,16), (3,4), (10,12) etc. are issued. The person who gets (4,10) will go to row 4 and occupy the 10th seat. Thus the first number denotes the row and the second number, the seat. Which seat will the visitor with token (5,9) occupy? Can he go to 9th row and take the 5th seat? Do (9,5) and (5,9) refer to the same location? No, certainly! What can you say about the tokens (2,3), (6,3) and (10,3)?



Fig. 1.1

This is one example where a pair of numbers, written in a particular order, precisely indicates a location. Such a number pair is called an ordered pair of numbers. This notion is skillfully used to mathematize the concept of a "Relation".

1.3 Cartesian Product

Illustration 1

Let us consider the following two sets.

A is the set of 3 vegetables and B is the set of 4 fruits. That is,

A = {carrot, brinjal, ladies finger} and B = {apple, orange, grapes, strawberry}

What are the possible ways of choosing a vegetable with a fruit? (Fig.1.2)

Vegetables (A)	Fruits (B)
Carrot (c)	Apple (a)
Brinjal (b)	Orange (o)
Ladies finger (l)	Grapes (g)
	Strawberry (s)

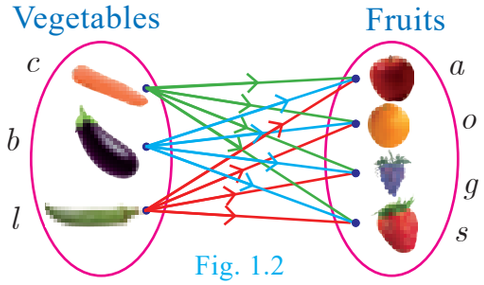


Fig. 1.2

We can select them in 12 distinct pairs as given below.

$(c, a), (c, o), (c, g), (c, s), (b, a), (b, o), (b, g), (b, s), (l, a), (l, o), (l,g), (l, s)$

This collection represents the cartesian product of the set of vegetables and set of fruits.

Definition

If A and B are two non-empty sets, then the set of all ordered pairs (a, b) such that $a \in A,\ b \in B$ is called the Cartesian Product of A and B, and is denoted by $A \times B$.

Thus, $A \times B = \{(a,b) \mid a \in A, b \in B\}$ (read as A cross B). Also note that A $\times$ $\phi = \phi$

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